

Problem 1. (LU Decomposition and Cholesky Decomposition)(25 points)

Let $\alpha, b \in \mathbb{R}^n$ and $A = I_n + \alpha\alpha^T$. You need to program the following methods to solve $Ax = b$:

- **The inverse method:** Use Gaussian elimination to find the inverse of A , and then compute $x = A^{-1}b$
- **LU decomposition:** First find the LU decomposition of A , then solve $Ly = b$ and $Ux = y$.
- **Cholesky Decomposition:** First find the Cholesky decomposition of A , then solve $Gy = b$ and $G^T x = y$.

In your homework, you are required to submit a time-consuming plot (one figure) of the given methods against the size of matrix A (i.e., n), where $n = 100, 200, \dots, 1000$.

Remarks:

- 1) Use the `randn` function to generate α and b .
- 2) Do not use the built-in Matlab functions `A\b`, `inv(A)`, `lu(A)` or `chol(A)`. Otherwise, you will lose points.